

1. A disadvantage of using bacterial cells to grow proteins is that _____.
 - a. They are simple to grow
 - b. Their media is inexpensive
 - c. They are most like human cells
 - d. They grow quickly
 - e. They do not correctly fold all proteins
2. Which of the following methods is used to detect proteins during downstream processing?
 - a. X-ray crystallography
 - b. Sonication
 - c. Polymerase chain reaction (PCR)
 - d. SDS-PAGE
 - e. Lyophilization
3. All the following methods are used for stabilizing proteins EXCEPT:
 - a. Keeping the protein cold
 - b. The addition of anti-foaming agents
 - c. The addition of protease inhibitors
 - d. Addition of DNase to the protein solution
 - e. Buffering of the pH
4. Inclusion bodies are clumps of insoluble proteins produced inside eukaryotic cells during upstream processing.
 - a. True
 - b. False
5. Large containers used for growing large cultures of cells under aerobic conditions are called:
 - a. Fermenters
 - b. Processors
 - c. Electroporators
 - d. Bioreactors
 - e. Sequencers

6. Cheese manufacturing involves enzymatic or microbial processing of the milk protein _____ to form a coagulated mixture called curd.
- ☐ a. Lactic acid
 - ☐ b. Chymosin
 - ☐ c. Casein
 - ☐ d. Malt
 - ☐ e. Rennin
7. Dialysis could be used to separate proteins from each other based on solubility.
- ☐ a. True
 - ☐ b. False
8. Cells must be lysed to release the protein of interest if _____.
- ☐ a. The protein is positively charged
 - ☐ b. The protein is hydrophilic
 - ☐ c. The protein is hydrophobic
 - ☐ d. The protein is secreted into the media
 - ☐ e. The protein is intracellular
9. What is the name for the technique that uses one antibody to capture a protein and a second, enzyme-conjugated antibody to detect the protein?
- ☐ a. Enzyme-linked immunosorbent assay (ELISA)
 - ☐ b. Lyophilization
 - ☐ c. X-Ray crystallography
 - ☐ d. Isoelectric focusing
 - ☐ e. SDS-PAGE
10. Proteins on SDS-PAGE gels are visualized by staining with Ethidium bromide.
- ☐ a. False
 - ☐ b. True
11. In hydrophobic interactions chromatography, proteins with the highest hydrophobicity will be eluted first.
- ☐ a. True

- b. False

12. Fungi are prokaryotic and capable of posttranslational modifications to allow proper folding of proteins.

- a. False
- b. True

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
E	D	B	B	D	C	B	E	A	A	B	A